Aakash
Medical | IIT-JEE | Foundations
Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **Fortnightly Test for NEET-2027 (XII-Passed)_FT01C** **Time : 180 Min.**

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line
Chemistry: Some Basic Concepts of Chemistry
Botany: Cell: The Unit of Life
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules-I: (Upto polysaccharides)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A body having initial velocity u is uniformly accelerated with an acceleration a for time t, then the average velocity over the time t is
(1) $u + at$ (2) $\frac{u+at}{2}$
(3) $\frac{2u+at}{2}$ (4) $\frac{u+at^2}{2}$

2. Two balls P and Q are projected in vertically upward direction with velocities 40 m s^{-1} and 80 m s^{-1} respectively. The ratio of distances moved by them during first second of their downward journey will be
(1) 1 : 2 (2) 2 : 1
(3) 1 : 1 (4) 1 : 4

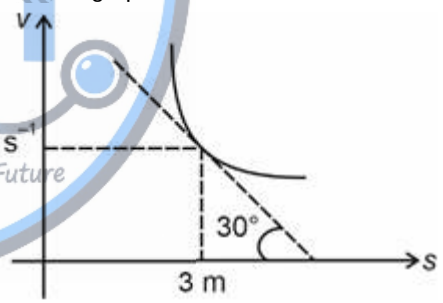
3. The acceleration of a particle moving along a straight line is given by
$a = (4v^2 - 20v + 16)$ m/s^2 where v is speed. When acceleration is zero, the speed achieved by the particle would be
(1) 4 m/s (2) 1 m/s
(3) 8 m/s (4) Both (1) and (2)

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4. A car is moving with uniform acceleration along a straight line between two points x and y which are distance $2d$ apart. Its speed at x and y are 1 m s^{-1} and 7 m s^{-1}, respectively then its speed at distance d from x is
(1) 10 m s^{-1} (2) 5 m s^{-1}
(3) 4 m s^{-1} (4) 3 m s^{-1}

5. A particle starts from rest and moving with constant acceleration 3 m/s^2. Distance travelled by the particle in 4th second of its journey is
(1) 24 m (2) 37.5 m
(3) 10.5 m (4) 15 m

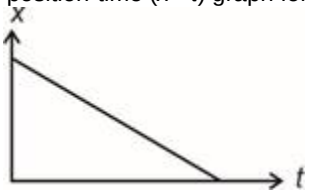
6. A particle is moving along a straight line whose velocity-displacement graph is as shown.



The magnitude of acceleration of the particle when its displacement is 3 m, is
(1) $4\sqrt{3}$ m s^{-2} (2) $\frac{\sqrt{3}}{4}$ m s^{-2}
(3) $\frac{4}{\sqrt{3}}$ m s^{-2} (4) $\sqrt{3}$ m s^{-2}

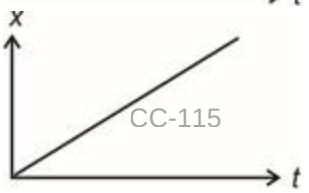
7. When a ball is thrown vertically up with velocity v_0, it reaches a maximum height of h_0. If one wishes to double the maximum height, then the ball should be thrown with a velocity of
(1) $\sqrt{3}v_0$ (2) $\frac{v_0}{\sqrt{2}}$
(3) $\sqrt{2}v_0$ (4) $\frac{v_0}{\sqrt{3}}$

8. The position-time (x - t) graph for positive acceleration is

(1) 

(2) 

(3) 

(4) 

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9. If a freely falling body starting from rest travels a distance in last second of its journey which is equal to the distance travelled by it in the first three second, then time of flight of the body is

(1) 3 second (2) 5 second
(3) 7 second (4) 4 second

10. A body dropped from top of a tower falls through 40 m during the last two seconds of its fall. The height of tower is (g = 10 m/s^2)

(1) 60 m (2) 45 m
(3) 80 m (4) 50 m

11. The position of a particle related to time is given by $x = (5t^2 - 4t)$ metre, where t is in second. The time at which velocity will become zero, is

(1) 0.8 s (2) 4 s
(3) 0.15 s (4) 0.4 s

12. Two trains A and B, move on parallel tracks. Train A moves towards east at 72 km/h and train B moves towards west at 54 km/h. A bird flies between them at 90 km/h, starting from train A when they are 1.75 km apart and touching each train alternately until they collide. The total distance flown by the bird will be

(1) 1.25 km (2) 1 km
(3) 1.5 km (4) Infinite

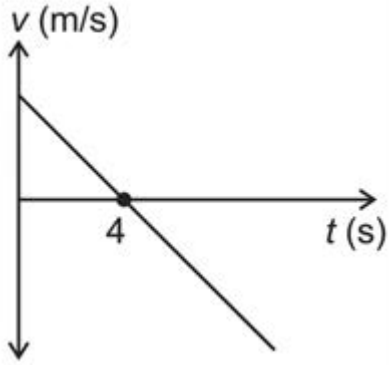
13. For a particle undergoing uniform motion along a straight line, the correct statement is:

(A) Its position-time graph is a straight line.
(B) Its velocity-time graph is a straight line.
(C) Its acceleration-time graph will be parabolic.

Choose the correct answer among the options given below.

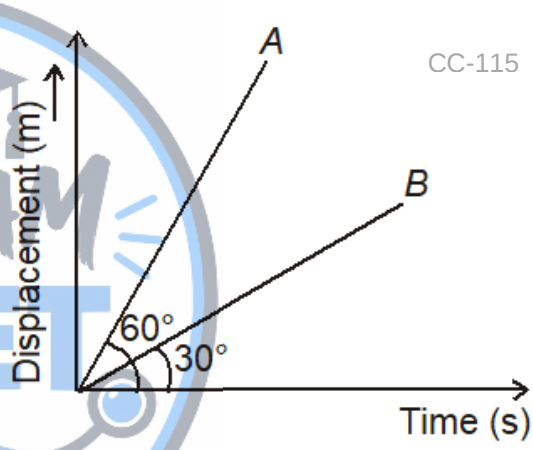
(1) (A) only (2) (B) Only
(3) Only (A) and (B) (4) All (A), (B) and (C)

14. The given graph is for a particle projected upwards from a height of 40 m. The initial velocity of the particle is ($g = 10\ \text{m/s}^2$)



(1) 20 m/s (2) 40 m/s
(3) 80 m/s (4) 10 m/s

15. The ratio of velocities represented by lines A and B $\left[\frac{V_A}{V_B}\right]$ in the graph below is



(1) 1 : 1 (2) 1 : 2
(3) 3 : 1 (4) 1 : 3

16. A particle is moving along x-axis with its position (x) Varying with time (t) as $x = (t^3 + t^2)$m. The ratio of magnitude of instantaneous velocity at t = 2 s to magnitude of average velocity between t = 0 to t = 2 s is

(1) $\frac{4}{3}$ (2) $\frac{8}{3}$
(3) $\frac{16}{7}$ (4) 4

17. A ball is thrown vertically upward with speed u. It passes through same point at 3 second and 7 second from the start. The speed of projection is

(1) 30 m s^{-1} (2) 70 m s^{-1}
(3) 100 m s^{-1} (4) 50 m s^{-1}

18. The circular scale of a screw gauge has 100 equal divisions. When it is given 2 complete rotations, it moves to 2 mm. The least count of screw gauge is

(1) 0.01 mm (2) 0.01 cm
(3) 0.04 mm (4) 0.005 cm

19. The value of the product of two measured quantities 414.4 and 94 upto correct significant figures is

(1) 3.9×10^4 (2) 38953.6
(3) 3.6×10^3 (4) 36853.5

20. Dimensional formula for electric charge is
(1) $[A^0T]$ (2) $[AT^{-1}]$
(3) $[A^{-1}T]$ (4) $[AT]$

21. The quantity having both unit and dimensions, is
(1) Velocity (2) Plane angle
(3) Strain (4) Specific gravity

22. Time period of a simple pendulum is $T = 2\pi\sqrt{\frac{l}{g}}$. If percentage error in measurement of length and acceleration due to gravity is 2% and 4% respectively, then maximum percentage error in time period is
(1) 1% (2) 2%
(3) 3% (4) 4%

23. 10 N of force is to be converted in a new system of units in which unit of mass is 10 kg, unit of length is 10 m and unit of time is 10 s. The numerical value of given force in new system of units is
(1) 10 (2) 100
(3) 1000 (4) 1

24. Number of insignificant zeroes in the value 0.0023400 are
(1) 3 (2) 5
(3) 6 (4) 2

25. The smallest value that can be measured by the measuring instrument is called the
(1) Absolute error (2) Relative error
(3) Percentage error (4) Least count

26. In an experiment four quantities *a*, *b*, *c* and *d* are measured with percentage error 2%, 3% 1% and 0.5% respectively. A quantity *Q* is defined as $Q = \frac{a\sqrt{b}}{c^{3/2}d^4}$. Maximum percentage error in the calculation of *Q* will be
(1) 5% (2) 6%
(3) 7% (4) 9%

27. The quantity in SI system whose unit is not treated as a fundamental unit is
(1) Time (2) Mass
(3) Velocity (4) Length

28. A small metallic ball moves through a fluid with a speed *v* in downward direction, experiences a force *F* which acts in upwards direction. Experimentally, the magnitude of this force depends on radius '*r*' of ball, velocity '*v*' of ball and viscosity 'η' of fluid. The correct relation between *F*, *r*, *v* and η is given by
(1) $F \propto \eta r v$ (2) $F \propto \eta^2 r v$
(3) $F \propto \eta r^{-2} v$ (4) $F \propto \eta^{-1} r^{-1} v^{-1}$

29. A physical equation is given by $v = \alpha[4 - e^{-\beta x t}]$ where *v* is speed, *x* is displacement and *t* is time. Dimensional formula of β^2 will be
(1) $[MLT^{-2}]$ (2) $[M^{-1}T^{-2}L^3]$
(3) $[M^0L^{-2}T^{-2}]$ (4) $[M^{-1}L^{-2}T^1]$

30. The fundamental quantity which has same dimension in dimensional formula of velocity gradient and pressure gradient is
(1) Mass (2) Length
(3) Time (4) None of these

31. Which of the following formula is dimensionally correct? (Symbols have their usual meanings)
(1) $F = \frac{2mv^2}{r^2}$ (2) $F = \frac{mv^2}{r}$
(3) $F = m^2a$ (4) $F = \frac{mv}{r^2}$

32. The density of water is 1 g/cc. The numerical value of density of water in new unit kg/m^3 is
(1) 1 (2) 10
(3) 10^2 (4) 10^3

33. A new unit of length is equal to 10 m. The area of 5 m^2 expressed in terms of new unit has magnitude
(1) 0.05 (2) 0.50
(3) 5.05 (4) 5.00

34. **Assertion (A) :** The unit of plane angle and solid angle is same.
Reason (R) : The unit of plane angle and solid angle is radian.
(1) Both assertion and reason are true and the reason is the correct explanation of the assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion and reason both are false

35. Given below are two statements:
Statement-I: A choice of change in different units does not change the number of significant figures in a measurement.
Statement-II: when two physical quantities are multiplied, there should be as many decimal places retained in the final result as are there in the number with the least decimal places.
In light of above statements, choose the most appropriate answer from the options given below
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

36. Match the column (I) with column (II) and choose the correct option.

Column (I)	Column (II)
(i) Area under velocity-time graph	(a) Gives velocity
(ii) Area under acceleration- time graph	(b) Gives acceleration
(iii) Slope of velocity-time graph	(c) Gives displacement
(iv) Slope of position-time graph	(d) Gives change in velocity

(1) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)
(2) (i)-(c),(ii)-(d), (iii)-(b), (iv)-(a)
(3) (i)-(b),(ii)-(c), (iii)-(d), (iv)-(a)
(4) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)

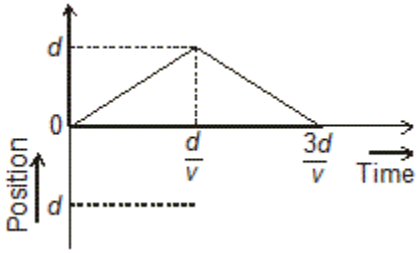
37. Two parallel rail tracks run north-south. Train A moves north with uniform speed 15 m/s and train B moves south with uniform speed 25 m/s. Then match the column-I with column-II

Column-I	Column-II
A. Magnitude of velocity of train *B* w.r.t. train *A*	p. Zero
B. Magnitude of velocity of ground w.r.t. train *B*	q. 40 m/s
C. Magnitude of velocity of ground w.r.t. train *A*	r. 15 m/s
D. Magnitude of acceleration of train *A* w.r.t. train *B*	s. 25 m/s

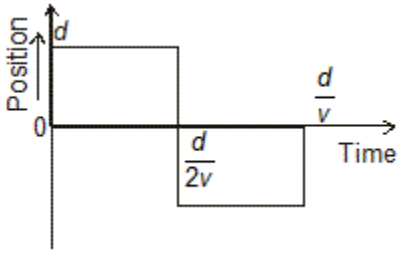
(1) A(p), B(q), C(r), D(s) (2) A(q), B(s), C(r), D(p)
(3) A(q), B(p), C(s), D(r) (4) A(r), B(s), C(p), D(q)

38. A person travelled a distance d with constant velocity v and return back to initial position with constant velocity $2v$. The position-time graph of the above situation is

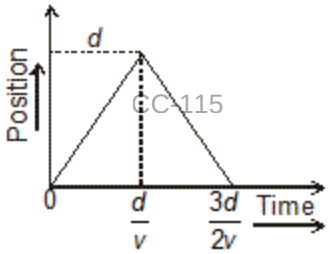
(1)



(2)



(3)



(4) Undefined

39. Read the following statements carefully and mark the correct option.
Statement(A): Dimensionless quantities must have no unit.
Statement(B): Unitless quantities must have no dimension.
Statement(C): Dimensionless quantities may have unit.
(1) Statements (A) and (B) are correct
(2) Statements (B) and (C) are correct
(3) All statements (A), (B) and (C) are correct
(4) Only statement (A) is correct

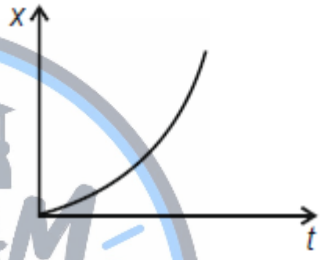
40. Consider the following statements and tick the correct option.
Statement A: If a body is moving under constant acceleration, then its direction of motion remains unchanged.
Statement B: If at any instant, acceleration of any object is zero, then the object should have zero velocity at that instant.
Statement C: Magnitude of instantaneous velocity is always equal to instantaneous speed.
(1) A and B are true while C is false
(2) A and C are true while B is false
(3) A and B are false while C is true
(4) A, B and C all are false

41. In the following question, a statement of **Assertion (A)** is followed by a statement of **Reason (R).**
Assertion (A) : If two particles, moving along straight line with constant velocities have to meet, the relative velocity must be along the line joining the two particles.
Reason (R) : Relative motion means motion of one particle as viewed from the other particle.
(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)
(2) Both (A) and (R) are true and the (R) is not the correct explanation of the (A)
(3) (A) is true statement but (R) is false
(4) Both (A) and (R) are false statements

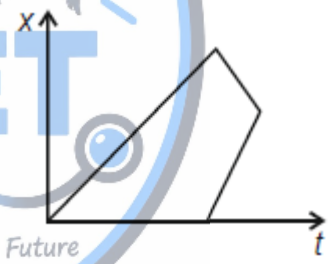
42. $\int_0^{\pi/4} \cos 2x \, dx =$
(1) 1 (2) 2
(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

43. Which of the following position-time graphs can possibly represent one dimensional motion of a particle?

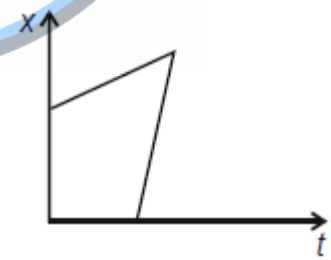
(1)



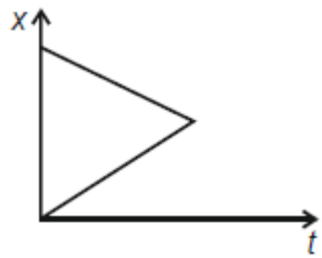
(2)



(3)



(4)



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44. In the figure given below, a particle's position is marked at equal time intervals. Select the correct statement.



(1) Speed is first increasing and then decreasing
(2) Speed is first decreasing and then increasing
(3) Speed is constant everywhere
(4) Speed is always decreasing

45. The following chart shows position *x* with time *t* of a particle

x(m)	*t*(s)
1	0
–1	2
3	4
–3	6

The magnitude of average velocity of the particle in time (0-6) s is

(1) 4 m/s
(2) $\frac{2}{3}$ m/s
(3) $\frac{4}{3}$ m/s
(4) $\frac{8}{3}$ m/s

CHEMISTRY

46. The correctly reported answer of addition of 1.234 and 1.1 will have significant figures

(1) One
(2) Two
(3) Three
(4) Four

47. Number of electrons present in 9.6 g of SO_4^{2-} ion is

(1) 2 N_A
(2) 5 N_A
(3) $\frac{5}{N_A}$
(4) $\frac{6}{N_A}$

48. Consider the following statements:
Statement I: Precision refers to the agreement of a particular value to the true value of the result.
Statement II: Accuracy refers to closeness of various measurements for the same quantity.
In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

49. Which of the given pairs has the compounds with same empirical formula?

(1) Benzene and cyclohexane
(2) Ethyne and Propyne
(3) Benzene and Ethyne
(4) Propyne and Cyclohexane

50. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion(A): Equal volumes of ammonia and carbon dioxide samples at STP contain same mass.
Reason(R): Equal volumes of all gases at same temperature and pressure should contain equal number of molecules.
In the light of the above statements, choose the correct answer.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is incorrect
(4) (A) is incorrect but (R) is correct

51. If percentage of Zn (atomic mass = 65.38 u) in a biomolecule is 0.75% by weight, then the minimum molar mass (in g mol^{-1}) of biomolecule will be

(1) 10243.45
(2) 9483.29
(3) 8012.12
(4) 8717.33

52. Match the List I containing multiples with their respective prefixes given in List II.

	List I		**List II**
(a)	10^{-15}	(i)	yocto
(b)	10^{-12}	(ii)	zepto
(c)	10^{-21}	(iii)	femto
(d)	10^{-24}	(iv)	pico

Choose the correct match:

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

53. An element E exists in nature in the form of two isotopic forms with atomic masses 12 u and 13 u. If the average atomic mass of E is 12.2 u then the percentage abundances of the two isotopes is

(1) 10% and 90%
(2) 30% and 70%
(3) 20% and 80%
(4) 25% and 75%

54. Consider the following statements

(a) 1 amu = 1.66056×10^{-27} kg

(b) 1 mole of atomic H contains 6.022×10^{-23} hydrogen atoms

(c) SI unit of luminous intensity is candela

The correct statements are

(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

55. Consider the following statements about Dalton's atomic theory.
a. Matter consists of indivisible atoms.
b. Atoms of different elements differ in mass.
c. Compounds are formed when atoms of different elements combine in a fixed ratio.
d. Atoms are neither created nor destroyed in a chemical reaction.
Correct statements are

(1) a and b only
(2) b and c only
(3) c and d only
(4) a, b, c and d

56. Among the following for which formula mass is applicable is

(1) NaCl
(2) CH_4
(3) Cl_2
(4) H_2

57. Find out the correct statement(s)

(a) Mass of 3.01×10^{23} molecules of methane is 8 gram
(b) A pure compound has always a fixed proportion of masses of its constituents
(c) 18 gram of water and 18 gram of ice will contain same number of molecules.

(1) (a), (b) and (c) (2) (a) and (c) only
(3) (a) and (b) only (4) (b) and (c) only

58. Consider the following statements

Statement-I: Gay Lussac's Law of Gaseous Volumes : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided that all gases are at same temperature and pressure.

Statement-II: The volumes of $H_2(g)$ and $O_2(g)$ which combine together to form water bear a simple ratio of 2 : 1

In the light of above statements, choose the correct answer from the options given below.

(1) Statement I is false but statement II is true
(2) Both statements I and II are true
(3) Both statements I and II are false
(4) Statement I is true but statement II is false

59. Suppose the elements A and B combine to form two compounds AB and AB_2. If 0.1 mole of AB weighs 6 g and 0.5 mole of AB_2 weighs 40 g, then atomic mass of A and B respectively are

(1) 20 u and 40 u (2) 40 u and 20 u
(3) 80 u and 40 u (4) 40 u and 80 u

60. If the mass ratio of He to N_2 in a container is 2 : 1 then the ratio of their moles will be

(1) 2 : 1 (2) 4 : 1
(3) 7 : 1 (4) 14 : 1

61. If total number of electrons in a sample of CO_2 is 4.4×10^{26} then total number of atoms present in the sample is

(1) 6×10^{26} (2) 6×10^{25}
(3) 2×10^{26} (4) 4.5×10^{26}

62. The crystalline salt, $Na_2SO_4.xH_2O$ on heating loses 55.9% of its weight. The formula of the crystalline salt is

(1) $Na_2SO_4.5H_2O$ (2) $Na_2SO_4.7H_2O$
(3) $Na_2SO_4.10H_2O$ (4) $Na_2SO_4.H_2O$

63. Match Column I with column II

Column I (Quantity)	Column II (% of oxygen)
a. Calcium carbonate	(i) 28.6
b. Sodium hydroxide	(ii) 40
c. Quick lime	(iii) 48
d. Glucose	(iv) 53.3

Choose the most appropriate match from the options given below:

(1) a-(ii), b-(i), c-(iv), d-(iii) (2) a-(iii), b-(i), c-(ii), d-(iv)
(3) a-(iii), b-(iv), c-(ii), d-(i) (4) a-(iii), b-(ii), c-(i), d-(iv)

64. Consider the following statements:

(i). Number of atoms in $80\ \mathrm{u}$ of calcium is 2
(ii). Number of atoms in $2240\ \mathrm{mL}$ of CO_2 at STP is 6.022×10^{22}
(iii). A sample of drinking water contains $10\ \mathrm{ppm}$ (by mass) of impurities. The percentage by mass of impurities in the sample is 10^{-3}

Choose the correct statement(s)

(1) (i) & (ii) only (2) (i) only
(3) (ii) & (iii) only (4) (i) & (iii) only

65. Which one of the following has maximum number of atoms?

(1) 1g of C(s) (2) 1g of $O_2(g)$
(3) 1g of $N_2(g)$ (4) 1g of $H_2(g)$

66. Mass of 80% pure $CaCO_3$ required to neutralise 100 mL of 0.5 M HCl solution will be

(1) 2.06 g (2) 3.12 g
(3) 4.15 g (4) 4.50 g

67. For the given reaction

$P_4 + 5\ O_2 \rightarrow P_4O_{10}$

If 31 g of P_4 is reacted with excess of oxygen and the percentage yield is 60%, then the mass of product formed is

(1) 42.6 g (2) 12.8 g
(3) 71 g (4) 28.4 g

68. Consider the given reactions:

(i) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

(ii) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

Mass of NO_2 produced from 0.5 mol of N_2 is

(1) 92 g (2) 72 g
(3) 69 g (4) 46 g

69. Consider the combustion reaction of 8 g of methane with 22.4 L oxygen gas at STP.

Statement I: Oxygen will get completely consumed.

Statement II: If the mass of both reactants is reduced to half, then the mass of CO_2 obtained will also get reduced to half.

In the light of above statements, choose the **correct** option.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Statement I is correct but Statement II is incorrect

70. One litre of CO_2 is passed through red hot coke. The volume becomes 1.4 litres at same temperature and pressure. The composition of products is

(1) 0.8 litre of CO_2 and 0.6 litre of CO
(2) 0.7 litre of CO_2 and 0.7 litre of CO
(3) 0.6 litre of CO_2 and 0.8 litre of CO
(4) 0.4 litre of CO_2 and 1.0 litre of CO

71. Volume of $CO_2(g)$ obtained at STP by complete decomposition of 0.1 mol $NaHCO_3$ is

(1) 2 L (2) 1.12 L
(3) 2.24 L (4) 1.5 L

72. 10 L Hydrogen gas and 20 L Chlorine gas combine in suitable conditions to produce hydrogen chloride gas. The composition of resultant gaseous mixture is

(1) 5 L Chlorine and 25 L Hydrogen Chloride
(2) 5 L Hydrogen, 10 L Chlorine and 10 L Hydrogen Chloride
(3) 10 L Hydrogen and 20 L Hydrogen Chloride
(4) 10 L Chlorine and 20 L Hydrogen Chloride

73. The amount of magnesium required to produce 112 ml of H_2 at STP by treatment with dilute H_2SO_4 will be

(1) 2.1 g (2) 0.12 g
(3) 1.2 g (4) 0.21 g

74. What is the mass of glucose required to produce 44 g of CO_2, on complete combustion?

(1) 30 g (2) 45 g
(3) 60 g (4) 22 g

75. Which of the following pairs of compounds follow law of multiple proportions?
(1) H_2O and Na_2O (2) CO_2 and CO
(3) CO_2 and NO_2 (4) BaO and BeO

76. If mole fraction of solvent in aq. H_2SO_4 solution is 0.85 then number of moles of solute present in 2 kg of solvent is
(1) 6.66 (2) 5.66
(3) 0.15 (4) 19.6

77. Mass percentage of solute in 1 molal aqueous glucose solution is approximately
(1) 20.0% (2) 15.2%
(3) 22.0% (4) 25.8%

78. The mole fraction of the solute in 4.5 m aqueous solution is approximately
(1) $\frac{7}{48}$ (2) $\frac{3}{40}$
(3) $\frac{6}{29}$ (4) $\frac{4}{58}$

79. If 3.01×10^{23} molecules of sugar are present in 24.5 mol of water then mole fraction of sugar in solution is
(1) 0.08 (2) 0.04
(3) 0.2 (4) 0.02

80. The mole fraction of methanol when equal masses of methanol and ethanol are mixed together is
(1) 39/23 (2) 12/17
(3) 23/39 (4) 17/12

81. Mass of glucose present in 2 kg of 1 molal aqueous solution of glucose is
(1) 305 g (2) 380 g
(3) 450 g (4) 480 g

82. 2 litres of 0.1 M HCl solution is mixed with another 1 litre of 0.2 M HCl solution. The molarity of the resulting solution will be
(1) 0.13 M (2) 0.23 M
(3) 0.31 M (4) 0.47 M

83. Number of urea molecules present in 400 mL of 2.5 M aqueous solution of urea is
(1) 3.01×10^{22} (2) 6.02×10^{23}
(3) 6.02×10^{26} (4) 3.01×10^{24}

84. 100 ml of 20.8% (w/v) $BaCl_2$ (aq) and 100 mL of 14.7% (w/v) H_2SO_4 (aq) solutions are mixed. Molarity of SO_4^{2-} ions in the resulting solution is [Molar mass of $BaCl_2$ = 208 g/mol]
(1) 0.5 M (2) 0.1 M
(3) Zero (4) 0.25 M

85. If the density of methanol is 0.793 Kg L^{-1}. The volume of methanol needed for making 3 L of 0.3 M aqueous solution is
(1) 36.32 mL (2) 28.34 mL
(3) 32.36 mL (4) 38.58 mL

86. If mole fraction of H_2SO_4 in aqueous solution is 0.1 then molality of H^+ and SO_4^{2-} ions respectively are (Assume 100% ionisation of H_2SO_4)
(1) 6.17 m, 6.17 m (2) 6.17 m, 12.34 m
(3) 12.34 m, 6.17 m (4) 12.34 m, 12.34 m

87. If drinking water contains 10 ppm (by mass) of chloroform, then the concentration in percentage by mass is
(1) 10^{-2} (2) 10^{-4}
(3) 10^{-1} (4) 10^{-3}

88. If the molality of the aqueous solution is 0.111, then the mole fraction of solute will be
(1) 0.02 (2) 0.002
(3) 0.2 (4) 0.1

89. If 0.3 mole of $BaCl_2$ is mixed with 0.4 moles of Na_3PO_4, then maximum number of moles of $Ba_3(PO_4)_2$ and NaCl formed are respectively
(1) 0.2 and 1.6 (2) 0.3 and 1.8
(3) 0.1 and 0.6 (4) 0.4 and 2.4

90. 1 mol of $KClO_3$ is thermally decomposed to KCl and O_2 and excess of aluminium is burnt in the gaseous product. How many moles of Al_2O_3 are formed?
(1) 1 (2) 2
(3) 1.5 (4) 3

BOTANY

91. Prokaryotic cells differ from eukaryotic cells as the former
(1) Have cell wall (2) Have cell membrane
(3) Contain ribosomes (4) Lack nuclear membrane

92. Who first explained that cells divided and new cells are formed from pre-existing cells?
(1) Schleiden (2) Schwann
(3) Rudolf Virchow (4) Leeuwenhoek

93. The endomembrane system is comprised of
(a) Golgi complex
(b) Lysosome
(c) Vacuole
(d) Endoplasmic reticulum
(e) Microfilaments
Select the **correct** ones:
(1) Only (a), (b) & (c) (2) Only (b), (c) & (d)
(3) Only (c), (d) & (e) (4) All except (e)

94. Given below are two statements : One is labelled as **Assertion** and the other is labelled as **Reason**.
Assertion : In plants, the concentration of ions is significantly higher in the vacuole than in the cytoplasm.
Reason : The tonoplast facilitates the transport of a number of ions against concentration gradient into the vacuole.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion and Reason are true but reason is not the correct explanation of the assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

95. Read the following statements and select the correct option
Statement A: The primary cell wall of a plant cell is capable of growth.
Statement B: The middle lamella glues the different neighbouring plant cells together.

(1) Only statement A is incorrect
(2) Only statement B is incorrect
(3) Both statement A and B are correct
(4) Both statement A and B are incorrect

96. Arrange the following cells in descending order of their size and choose the **correct** option.

Bacteria, Ostrich egg, *Mycoplasma*, Human RBC

(1) Ostrich egg > Bacteria > Human RBC > *Mycoplasma*
(2) Ostrich egg < Human RBC < Bacteria < *Mycoplasma*
(3) *Mycoplasma* < Bacteria < Human RBC < Ostrich egg
(4) Ostrich egg > Human RBC > Bacteria > *Mycoplasma*

97. Match the following.

(i) Plasmodesmata	(a) Nuclear matrix
(ii) Kinetochores	(b) Thick and tough glycocalyx
(iii) Nucleoplasm	(c) Disc shaped structure on centromere
(iv) Capsule	(d) Cytoplasmic connections

Choose the **correct** option.
(1) (i) – (d), (ii) – (a), (iii) – (c), (iv) – (b)
(2) (i) – (c), (ii) – (a), (iii) – (d), (iv) – (b)
(3) (i) – (d), (ii) – (c), (iii) – (a), (iv) – (b)
(4) (i) – (b), (ii) – (c), (iii) – (a), (iv) – (d)

98. The shape of nerve cells is
(1) Round and biconcave (2) Branched and long
(3) Amoeboid (4) Cuboidal

99. Satellite in chromosomes is
(1) Seen in telocentric chromosomes only
(2) A fragment beyond the secondary constriction
(3) Formed by primary constriction
(4) Devoid of DNA

100. Several ribosomes may attach to a single mRNA and form a chain called
(1) Polysome (2) Mesosome
(3) Chromosome (4) Peroxisome

101. An animal cell differs from a higher plant cell in possessing
(1) Vacuoles (2) Plastids
(3) Centrosomes (4) Nucleus

102. All of the following have a cell wall surrounding the cell membrane, **except** in
(1) *Mycoplasma* (2) Bacteria
(3) Blue-green algae (4) Plant cells

103. The major function of contractile vacuole is
(1) ATP synthesis (2) Nitrogen fixation
(3) Osmoregulation (4) Protein synthesis

104. The comma shaped bacterial cell is
(1) Bacillus (2) Coccus
(3) Spirillum (4) Vibrium

105. In prokaryotes, mesosomes help in all given processes, **except**
(1) Cell wall formation
(2) Respiration
(3) Hiding from host's immune system
(4) DNA Replication

106. Select the **odd** one w.r.t. function of cell wall in an organism.
(1) Maintenance of shape of cells
(2) Protection from mechanical injury
(3) Protection from the attack of pathogens
(4) Involvement in secretion and endocytosis processes

107. Which pair of structures are usually found in both plant and animal cells?
(1) Cell membrane and nucleolus
(2) Cell membrane and cell wall
(3) Nucleolus and chloroplast
(4) Nucleus and cell wall

108. Special membranous extension present in the form of vesicles, tubules and lamellae in Gram positive bacteria is called
(1) Glycocalyx (2) Mesosome
(3) Capsule (4) Slime layer

109. Which of the following structures lacks membrane?
A. Centriole
B. Flagella
C. Ribosome
D. Microbodies
E. Vacuoles
F. Lysosomes
(1) A & B (2) D & E
(3) A & C (4) D & F

110. Select the **incorrect** statement **w.r.t** lysosome
(1) Part of endomembrane system
(2) Lysosomal enzymes do not digest nucleic acids
(3) Formed by packaging process in golgi apparatus
(4) Rich in hydrolytic enzymes

111. Select the **incorrect** match.
(1) Mitochondria – Divide by fission
(2) Contractile vacuole – Excretion in Amoeba
(3) Cytoskeleton – Absent in eukaryotes
(4) Flagellum – Covered with plasma membrane in eukaryotes

112. Which of the following are three components of bacterial flagella?
(1) Filament, pili and fimbriae
(2) Hook, filament and axoneme
(3) Hook, filament and basal body
(4) Hook, pili and axoneme

113. Which one of the following is **incorrect** regarding ribosomes?
(1) Composed of rRNA and proteins
(2) Involved in protein synthesis
(3) Not surrounded by any membrane
(4) Eukaryotes have 80S ribosomes only

114. Live cells were first observed and described by
(1) M. Schleiden
(2) Robert Hooke
(3) Antonie Von Leeuwenhoek
(4) Robert Brown

115. Which of the following is **not** a function of the cell organelle that constitute an endomembrane system?
(1) Synthesis of ribosomal RNA
(2) Synthesis of steroidal hormones
(3) Osmoregulation and excretion
(4) Formation of glycoproteins and glycolipids

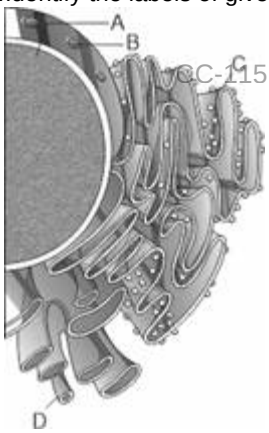
116. Select the **correct** match.
(1) Ribosomes – Non-membrane bound and made up of DNA and proteins
(2) Golgi apparatus – Important site for formation of glycolipids and glycoproteins
(3) Secondary wall – Thin and capable of extension growth of cells
(4) Endomembrane system – Functions not co-ordinated

117. In cyanobacteria, the membranous extensions into the cytoplasm, containing the pigments are called
(1) Mesosomes (2) Chromatophores
(3) Polysomes (4) Basal bodies

118. Number of inter doublet linkers associated with the axoneme of flagellum are
(1) 18 (2) 9
(3) 20 (4) 27

119. Select the incorrect match :
(1) Camillo Golgi – Golgi bodies
(2) Robert Brown – Nucleus
(3) Flemming – Chromatin
(4) George Palade – Mitochondria

120. Identify the labels of given diagram



Choose the **correct** statements w.r.t. given labels.
A. It contains loose and indistinct network of nucleoprotein fibres containing DNA and some basic proteins along with RNA.
B. It is formed due to the fusion of two membranes of nucleus.
C. It is single membrane bound organelle that is the major site for synthesis of lipoprotein and glycogen.
D. It is responsible for the formation of lipid and steroid.
(1) Only A, B and D (2) Only A, C and D
(3) Only A and B (4) Only C

121. Match the following columns and select the **correct** option

Column I		Column II
a. Centrosome	(i)	Made up of nine evenly spaced peripheral fibrils of tubulin, and forms the basal body of flagella
b. Centromere	(ii)	Is present in the interphase nucleus and represents highly extended and elaborate nucleoprotein fibres
c. Centrioles	(iii)	Usually contain two cylindrical structures which are surrounded by amorphous pericentriolar material
d. Chromatin	(iv)	Holds two chromatids of a chromosome together

(1) a-iii, b-i, c-iv, d-ii (2) a-iv, b-i, c-iii, d-ii
(3) a-iii, b-iv, c-i, d-ii (4) a-iii, b-ii, c-iv, d-i

122. Read the following statements and choose the **correct** option to fill in the blanks (a), (b) and (c).
(I) Chemical reactions occurring in the (a) keep the cell in living state.
(II) (b) is a membraneless structure which is absent in higher plants.
(III) (c) reported that animal cells have a thin outer layer.

	(a)	(b)	(c)
(1)	Nucleus	Flagella	Leeuwenhoek
(2)	Mitochondria	Ribosome	Schleiden
(3)	ER	Contractile vacuole	Rudolf Virchow
(4)	Cytoplasm	Centriole	Schwann

(1) (1) (2) (2)
(3) (3) (4) (4)

123. Centrioles differ from axoneme of cilia or flagella
(a) In number of microtubules present at periphery.
(b) In presence of radial spokes.
(c) In arrangement of microtubules.
(d) In containing proteinaceous central hub.
(e) In being surrounded by double membrane.
Choose the **correct** statements.
(1) (a), (c) and (d) only (2) (a), (b), (c) and (d)
(3) (a), (b) and (c) only (4) (b), (c), (d) and (e)

124. Match the following columns and select the **correct** option.

Column I		Column II
a. Metacentric chromosome	(i)	Centromere is located near one end, forming one very short and one very long arm
b. Sub-metacentric chromosome	(ii)	Centromere is located at the terminal end of the chromosome
c. Acrocentric chromosome	(iii)	Centromere is slightly away from the middle, forming one shorter and one longer arm.
d. Telocentric chromosome	(iv)	Centromere is located at the middle, forming two equal arms

(1) a(iv), b(i), c(iii), d(ii) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(iv), c(ii), d(i)

125. Mitochondria and chloroplast are not similar as the former does not
(1) Produce ATP
(2) Have double membrane envelope
(3) Have ds circular DNA
(4) Convert light energy into chemical energy

126. Match the column I and column II and select the **correct** option.

	Column I		Column II
a.	Oil storage	(i)	Aleuroplast
b.	Oxysomes	(ii)	Mitochondria
c.	Protein storage	(iii)	Elaioplast
d.	Light reaction	(Iv)	Chloroplast

(1) a(i), b(ii), c(iv),d(iii) (2) a(iii), b(ii), c(i),d(iv)
(3) a(ii), b(i), c(iii),d(iv) (4) a(iii), b(i), c(ii),d(iv)

127. Read the given statements and select **correct** option.
A: The *cis* and the *trans* face of Golgi apparatus are entirely different and are not interconnected.
B: Golgi apparatus remains in close association with ER.
C: Cisternae in Golgi apparatus are stacked parallel to each other.
(1) Only statement A is incorrect.
(2) Statements A and C are correct.
(3) All statements A, B and C are correct.
(4) Statements A and B are incorrect.

128. 'Bacterial cells have chemically complex cell envelope.' Choose the **incorrect** statement w.r.t. the bacterial cell envelope.
(1) The outermost layer is called glycocalyx
(2) All the layers together act as single protective unit
(3) Capsule is a thin glycocalyx layer that entirely lacks polysaccharides
(4) On the basis of difference in cell envelope and the manner in which they respond to the staining procedure developed by Gram, bacteria are classified into two groups

129. Which of the given forms of chromosome(s) possess both p- and q- arm?
(A) Metacentric chromosomes
(B) Sub-metacentric chromosome
(C) Acrocentric chromosome
(D) Telocentric chromosome
(1) Both (A) and (B) (2) Only (A)
(3) Both (B) and (C) (4) Only (D)

130. Select the **correct** option stating **True** (T) and **False**(F).
A. The fungal cell wall is generally made up of cellulose.
B. Plasmodesmata connect the cytoplasm of neighbouring animal cells.
C. Na^+/K^+ pump in animal cells is an energy dependent process.

	A	B	C
(1)	T	F	T
(2)	F	F	T
(3)	T	F	F
(4)	F	T	T

(1) (1) (2) (2)
(3) (3) (4) (4)

131. Identify the **incorrect** match w.r.t. plasma membrane.
(1) Lipid component - Mainly comprised of phospholipids
(2) Peripheral proteins - Lie on the surface of membrane
(3) Integral protein - Partially or totally buried in membrane
(4) Non-polar tail of lipid - Projected towards extacellular aqueous environment
(1) (1) (2) (2)
(3) (3) (4) (4)

132. In plants, the middle lamella present between the adjacent cells is mainly composed of
(1) Cellulose (2) Hemicellulose
(3) Calcium Carbonate (4) Calcium Pectate

133. The fimbriae present in bacteria are
(1) Elongated tubular structures of lipids
(2) Involved in movement of bacteria
(3) Small bristle-like fibres sprouting out of the cell
(4) Involved in photosynthesis

134. In human beings, the membrane of the erythrocyte has approximately _(A)_ protein and _(B)_ lipids.
Select the option that **correctly** fills (A) and (B) respectively.
(1) 40% and 52% (2) 52% and 40%
(3) 10% and 60% (4) 60% and 10%

135. The sausage shaped cell organelle, which has cristae, shows similarities in function with mesosomes. Choose the **correct** function.
(1) Digestion of macromolecules
(2) Steroid hormone synthesis
(3) Respiration
(4) Photosynthesis

ZOOLOGY

136. Which type of junctions in the epithelial tissues facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells for rapid transfer of ions or small molecules?
(1) Tight junctions (2) Adhering junctions
(3) Gap junctions (4) Desmosomes

137. Which of the following is **not** a correct statement for neuroglial cell?
(1) Neuroglial cells make up more than one half the volume of neural tissue in our body
(2) They protect and support the neurons.
(3) When these cells are suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane.
(4) They can multiply and divide in the mature nervous system.

138. Fusiform muscle fibres, that do not show striations and are involuntary will be found primarily in
(1) Limbs (2) Wall of stomach
(3) Wall of heart (4) Diaphragm

139. Brush bordered cuboidal epithelium is present in the lining of
(1) Proximal convoluted tubules
(2) Small intestine
(3) Stomach
(4) Smaller ducts of pancreas

140. Trapped dust particles are pushed out of respiratory tract by
(1) Squamous epithelium (2) Glandular epithelium
(3) Ciliated epithelium (4) Compound epithelium

CC-115

141. Read the following statements and choose the correct option.
Statement A: The matrix of the bone is present in the form of layers called lamellae.
Statement B: The matrix of bone is rich in calcium salts and collagen fibres.
(1) Only statement 'A' is true
(2) Only statement 'B' is true
(3) Both statements 'A' and 'B' are true
(4) Both statements 'A' and 'B' are false

142. The body of a simple organism like *Hydra* is made of
(1) A single cell
(2) Different types of cells
(3) Only single type of multiple cells
(4) Only two types of cells

143. Choose the **incorrect** match
(1) Skeletal muscle fibres – multinucleated
(2) Cardiac muscle fibres – branched in appearance
(3) Visceral muscle fibres – involuntary in nature
(4) Smooth muscle fibres – cylindrical in appearance

144. In general, the tissue that plays an active role in locomotion of the human body is
(1) Epithelial tissue (2) Connective tissue
(3) Muscular tissue (4) Neural tissue

145. Dense regular connective tissue includes
(1) Cartilage (2) Areolar tissue
(3) Adipose tissue (4) Ligament

146. Contractile tissue characterised by the presence of intercalated discs is found in
(1) Biceps (2) Heart
(3) Intercostal muscles (4) Diaphragm

147. All of the following are the characteristic features of lining of alveoli, **except**
(1) A single thin layer of flattened cells with irregular boundaries
(2) Cells are compactly packed with little intercellular matrix
(3) The cells are ciliated
(4) The nuclei of cells are flat and often lie at the centre of cells

148. Collagen fibres are secreted by
(1) Histiocytes (2) Fibroblasts
(3) Mast cells (4) Macrophages

149. An **odd** one w.r.t. specialized connective tissue is
(1) Bone (2) Tendon
(3) Cartilage (4) Blood

150. Select the odd one w.r.t. the presence of collagen fibres.
(1) Tendons
(2) Outer ear joints
(3) Beneath the skin
(4) Lymph present in lymph vessels

151. Cells of germinal epithelium are _____.
Select the correct option to fill in the blank.
(1) Cuboidal (2) Columnar
(3) Ciliated (4) Squamous

152. The type of tissue present at the outer ear joints is also found in the
(1) Tip of nose (2) Tendon
(3) Ligament (4) Cranial bone

153. Which tissue exerts the greatest control over the body's responsiveness to changing conditions?
(1) Muscle tissue (2) Neural tissue
(3) Connective tissue (4) Epithelial tissue

154. Structural syncytium is seen in muscles
(1) Of heart
(2) Present in blood vessels
(3) Constituting intestinal wall
(4) Associated with skeleton

155. Most abundant tissue in humans is
(1) Nervous tissue (2) Muscular tissue
(3) Connective tissue (4) Epithelial tissue

156. Complete the analogy.
Multicellular gland : Salivary gland : : Unicellular gland : _____
Choose the **correct** option.
(1) Goblet cell (2) Sebaceous gland
(3) Sweat gland (4) Gastric gland

157. Inner surface of hollow organs like bronchioles are lined by
(1) Simple flagellated columnar epithelium
(2) Simple ciliated columnar epithelium
(3) Compound ciliated columnar epithelium
(4) Simple ciliated squamous epithelium

158. Among the following, which secretion is directly poured into fluid bathing the gland?
(1) Sweat (2) Sebum
(3) Insulin (4) Saliva

159. Areolar tissue differs from adipose tissue as it
(1) Actively participates in inflammation and repair due to presence of mast cells
(2) Is a type of connective tissue that contains elastic and collagen fibres scattered randomly in its matrix.
(3) Is the abundantly distributed loose connective tissue and fills up the space between organs contrary to adipose tissue which primarily acts as a thermal insulator.
(4) Does not possess fat storing cells

160. Fibres of connective tissue do **not** provide
(1) Strength (2) Elasticity
(3) Excitability (4) Flexibility

161. A secondary metabolite which can also be used as a drug is
(1) Monoterpene (2) Diterpene
(3) Vinblastin (4) Carotenoid

162. In a molecule of glycogen, left end has
(1) Free aldehydic group representing reducing end
(2) Free carboxyl group representing C-terminal amino acid
(3) No free aldehydic group, representing non-reducing end
(4) Free ketonic group representing N-terminal amino acid

163. Starch and glycogen are common ____ found in plants and animals.
Select the correct option to fill in the blank.
(1) Polysaccharides (2) Disaccharides
(3) Oligosaccharides (4) Non reducing sugars

164. Comprehend the given statements w.r.t. starch and choose the incorrect one.
(1) It forms helical secondary structure.
(2) It can hold I_2 molecules in the helical portion.
(3) The starch-I_2 complex is red in colour.
(4) It is present as a store house of energy in plant tissues.

165. **Assertion :** Lipids though obtained in macromolecular fraction are not true macromolecules.
Reason : Lipids do not exceed molecular weight of 800 Da and are part of acid soluble fraction.
In the light of above statements, select the **correct** option.
(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true statement but Reason is false.
(4) Both Assertion and Reason are false statements.

166. For analysis of organic compounds, living tissue is grinded in
(1) Ash (2) Cl_3CCOOH
(3) CCl_4 (4) CH_3COOH

167. For ash analysis, living tissue is first dried and then burnt. The ash obtained subsequently contains
(1) Both organic and inorganic compounds
(2) Similar compounds which were present in acid insoluble pool while conducting compound analysis of a living tissue
(3) Only inorganic compounds/elements
(4) Only organic compounds

168. The element which forms 18.5% weight of human body is
(1) Hydrogen (2) Carbon
(3) Oxygen (4) Calcium

169. Biomolecule present in plant pulp that is used for making paper is
(1) Inulin (2) Cellulose
(3) Glycogen (4) Chitin

170. Structural homopolysaccharide formed from N-acetyl glucosamine is
(1) Cellulose (2) Chitin
(3) Starch (4) Collagen

171. Secondary metabolites like carotenoids and anthocyanins are classified as:
(1) Lectins (2) Essential oils
(3) Alkaloids (4) Pigments

172. The element present in greater quantity in the Earth's crust than human body is
(1) Carbon (2) Oxygen
(3) Calcium (4) Nitrogen

173. Inulin found in plant cell is a
(1) Lipid (2) Protein
(3) Polysaccharide (4) Vitamin

174. All of the given elements form higher per cent weight of Earth's crust in comparison to human body, **except**
(1) Sodium (2) Carbon
(3) Magnesium (4) Calcium

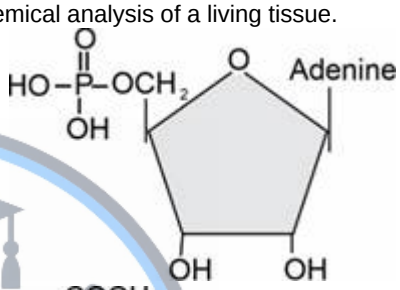
175. Inulin is a polymer of '__A__' and it can be used to measure GFR.
Choose the **correct** option for 'A'.
(1) Glucose (2) Fructose
(3) Galactose (4) Cellulose

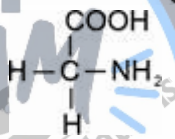
176. Most abundant structural polysaccharide in nature is a secondary metabolite, named as
(1) Glycogen (2) Starch
(3) Insulin (4) Cellulose

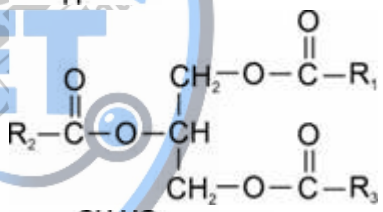
177. A dehydration reaction links one molecule of each X and Y to produce one molecule of Z. If the formula for each X and Y is $C_6H_{12}O_6$, then what is the formula of Z?
(1) $C_{12}H_{22}O_{11}$ (2) $C_{12}H_{24}O_{11}$
(3) $C_{12}H_{20}O_{10}$ (4) $C_{12}H_{24}O_{12}$

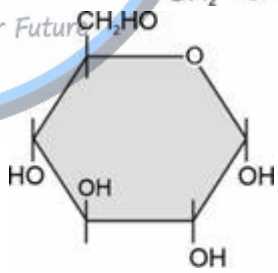
178. Complete the analogy by selecting the **correct** option w.r.t. the total number of glycosidic bond(s) in the given biomolecules.
Disaccharide : One : : Pentasaccharide : _____
(1) One (2) Two
(3) Three (4) Four

179. Among the biomolecules depicted below, choose the one that will be found in the retentate portion obtained upon the chemical analysis of a living tissue.

(1) 

(2) 

(3) 

(4) 

180. The biomolecule that **cannot** be hydrolysed further is
(1) Inulin (2) Insulin
(3) Lactose (4) Glucose

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